

Food and Nutrition

Lecture 21



Good nutrition helps fuel your body

Nutritional Requirements

Nutrient: *are the raw materials, which provide the essential matter and energy for life.*

- To scientists, **nutrition** is the study of:
 - The nutrients found in foods
 - The body's handling of nutrients to maintain health
- Nutrients (some of which provide energy for processes in the body) are substances that can promote:
 - **Growth**
 - **Maintenance**
 - **Repair**

Classes of Nutrients

- There are six classes of nutrients:

1. *carbohydrates,*
2. *fats (lipids) (20%),*
3. *proteins,*
4. *vitamins,*
5. *minerals, and*
6. *water (60%).*



Balanced Diet :

Balanced diet is a diet containing all the proximate principles of food in adequate and proportionate amounts; including essential amino acids and fatty acids for normal growth, activity, reproduction.

Adequate : Means the amount that supply the exact calories.

Proportionate amount : Means presence of appropriate amount of food stuffs.

Criteria of a balanced diet

- Proximate principles of food in adequate and proportionate amount.
- One third to half of protein.
- Fat should come from animal sources.
- Sufficient fruits and vegetables.
- Easily digestible, absorbable and assailable food.
- Easily available food.
- Certain amount of cellulose to promote peristalsis.

Factors considered to formulate a balanced diet

- Age
- Sex, height, weight
- Quality and quantity of food.
- Socio-economical value
- Seasonal factor
- Geographical
- Allergic condition

Energy Nutrients: Carbohydrates, Lipids and Proteins

- Used by body as source of energy. All these three nutrients are Carbon containing compounds.
- Proteins are **not preferred energy source**. They are **building blocks** to make our body.
- Body **does not get energy** from vitamins, minerals, water.

❑ Carbohydrate:

- Provide immediate energy through glucose.
- Store energy short-term as glycogen in liver and muscles.

❑ Protein:

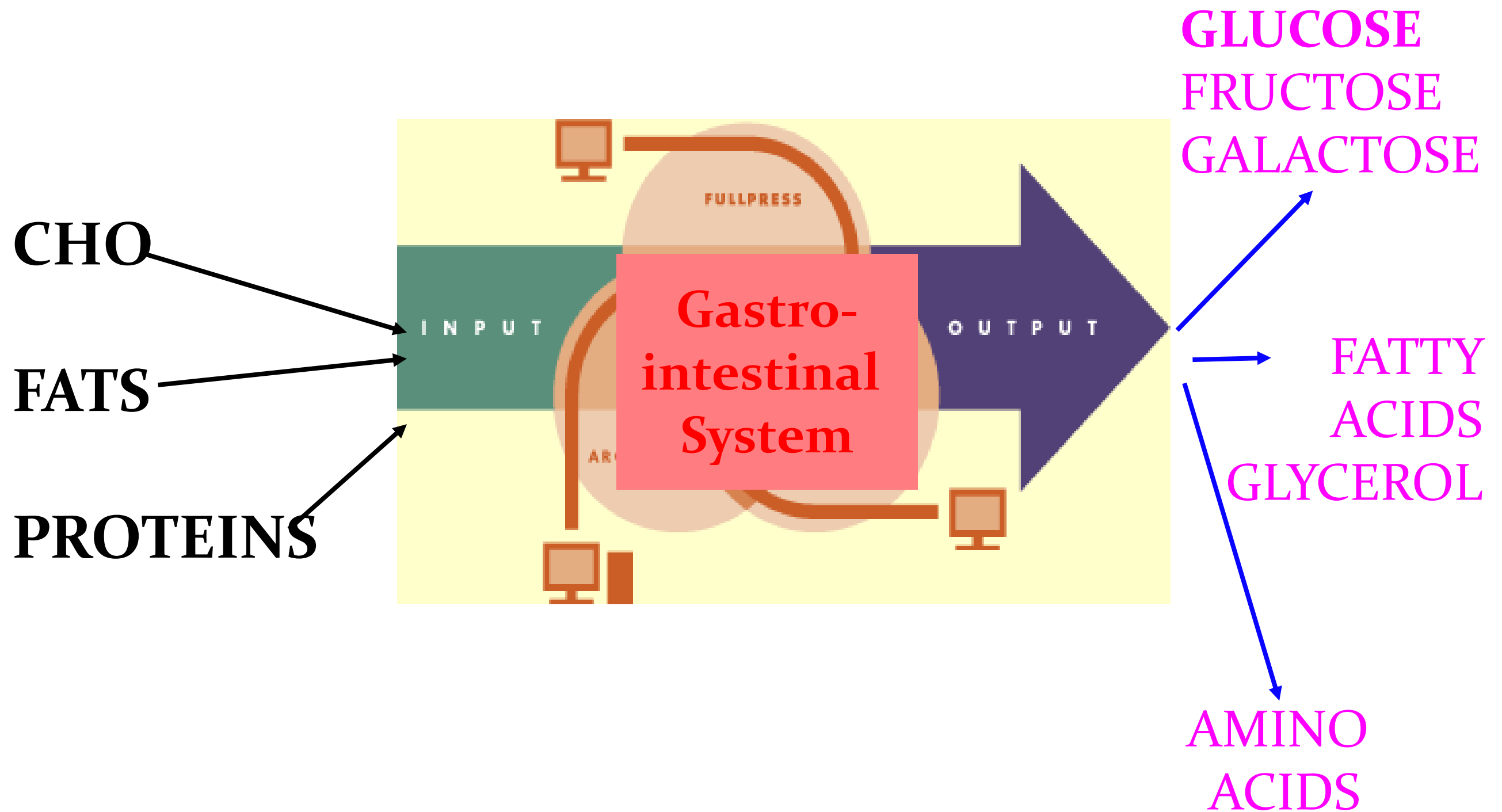
- Build and repair tissues such as muscles, skin, and organs.
- Form enzymes, hormones, and antibodies for body functions.

❑ Lipid:

- Store long-term energy in fat cells.
- Provide insulation and protect organs.

PRODUCTS OF DIGESTION

(Energy Nutrients)



Food as Fuel: Calories Count

□ Calorie value of food :

- Carbohydrate - 4.4 cal/g.
- Fat - 9.3 cal/g.
- Protein - 4.0 cal/g.
- A person would have to run for about 30 minutes, for example, to burn off the calories in a cheeseburger.

❖ Intake food(energy nutrients) > Energy needs : **stored in body fat.**

❖ Energy need > intake food : **burn body fat.**

Vitamins

A vitamin is an organic compound required as a vital nutrient in tiny amounts by an organism.

Vitamins are classified as:

1. Fat soluble Vitamins: A, D, E and K

- Taken with fat containing foods.
- They can remain stored in body fats.
- Not necessary to consume everyday.

2. Water soluble Vitamins: B complex and C.

- Cannot be stored.
- Need to take frequently/everyday.

In humans there are 13 vitamins:

i) 4 fat-soluble : A, D, E, and K and

ii) 9 water-soluble : 8 B vitamins (B-complex) and vitamin C

Vitamins

Water-soluble vitamins dissolve easily in water and, in general, are readily excreted from the body, to the degree that urinary output is a strong predictor of vitamin consumption. Because they are not readily stored, consistent daily intake is important. Many types of water-soluble vitamins are synthesized by bacteria.

Fat-soluble vitamins are absorbed through the intestinal tract with the help of lipids (fats). Because they are more likely to accumulate in the body, they are more likely to lead to **hypervitaminosis** than are water soluble vitamins. Fat soluble vitamin regulation is of particular significance in cystic fibrosis.

Minerals

- Some minerals are always required
- Fruits, vegetables and cereals are the chief sources of minerals
- Milk products supply the majority of calcium and phosphorous

Calcium:

- Increased amount of calcium is required by children, pregnant and lactating women
- The average intake should be about **800 mg per day**
- Best sources are milk and cheese. Other sources are eggs, green vegetables, oranges, nuts, beans, carrots eggs

Minerals

Phosphorous:

1. Children should take 1 g per day, adults 1.3 g and a pregnant or lactating woman 1.9 g
2. Sources of phosphorous are meat, fish, milk, cheese and eggs. Vegetable foods such as beans, oatmeal and lentils

Iron:

1. Deficiency leads to anemia
2. Sources are bread, meat and potatoes. Also liver, kidney, egg yolk, green peas, cabbage, carrots
3. Only 10% of iron is absorbed
4. Children up to 12 years require 15-20 mg. daily, adult 40 mg. and pregnant and lactating women 50 mg

Minerals

Iodine:

1. The daily requirement of iodine is 0.05 mg. and is obtained from water, vegetables and fish
2. Simple goitre results due to dietary deficiency

Magnesium:

1. The average daily consumption of about 0.2 is sufficient
2. Sources are meat, green vegetables and bread

Copper:

1. The daily requirement is 2 mg.
2. Sources are liver, cocoa, nuts

Sodium Chloride:

1. Usually 20 gm is consumed per day
2. The intake of salt is required only if abnormal quantities are lost e.g. severe exercise due to sweat; diarrhea
3. Heat stroke is prevented by ingestion of salt solutions instead of water

Water

- Water provides the medium in which all the body's reactions take place.
- It aids in the digestion and absorption of food.
- It is a medium of transport, through arteries and veins, within an organism.
- It even lubricates your joints and cushions organs such as the brain and spinal cord.
- It helps in the excretion of harmful by-products of metabolic processes.
- It aids in the regulation of heat loss.
- Key players when bonds are broken (by hydrolysis) or formed (by dehydration synthesis) in these molecules.

Body Mass Index (BMI)

- The **body mass index (BMI)** or **Quetelet index** is a value derived from the mass (weight) and height of an individual.
- The BMI is defined as the **body mass in Kilogram** divided by the **square of the body height in meter**, and is universally expressed in units of **kg/m²**, resulting from mass in Kilograms and height in meters.

Example: A person has body weight 68 Kg and 165 cm tall. Calculate the BMI of that person.

Calculation: Weight = 68 Kg; Height = 165 cm (1.65 m)

$$\therefore \text{BMI} = 68 \div (1.65)^2 = 24.98$$

Body Mass Index (BMI) count

BMI	Weight Status		
Below 18.5	Underweight	30.0 – 34.9	Obese Class I
18.5 – 24.9	Normal	35.0 – 39.9	Obese Class II
25.0 – 29.9	Overweight	40.0 and above	Obese Class III

The correlation between the BMI number and body fatness is fairly strong

however the correlation varies **by sex, race, and age.**

1. At the same BMI, **women tend to have more body fat** than men.
2. At the same BMI, older people, on average, **tend to have more body fat** than younger adults.
3. Highly trained athletes may have a high BMI because of increased muscularity rather than increased body fatness.

How To Calculate How Many Kilocalories You Need

1. Multiply the weight (in pounds)
by **10** if you are **not active physically**,
by **15** if you are **moderately active**, and
by **20** if you are **highly active**
2. **Subtract one of the following amounts** from the multiplication result

<i>Age:</i>	<i>25–34</i>	<i>Subtract:</i>	<i>0</i>
	35–44		100
	45–54		200
	55–64		300
	Over 65		400

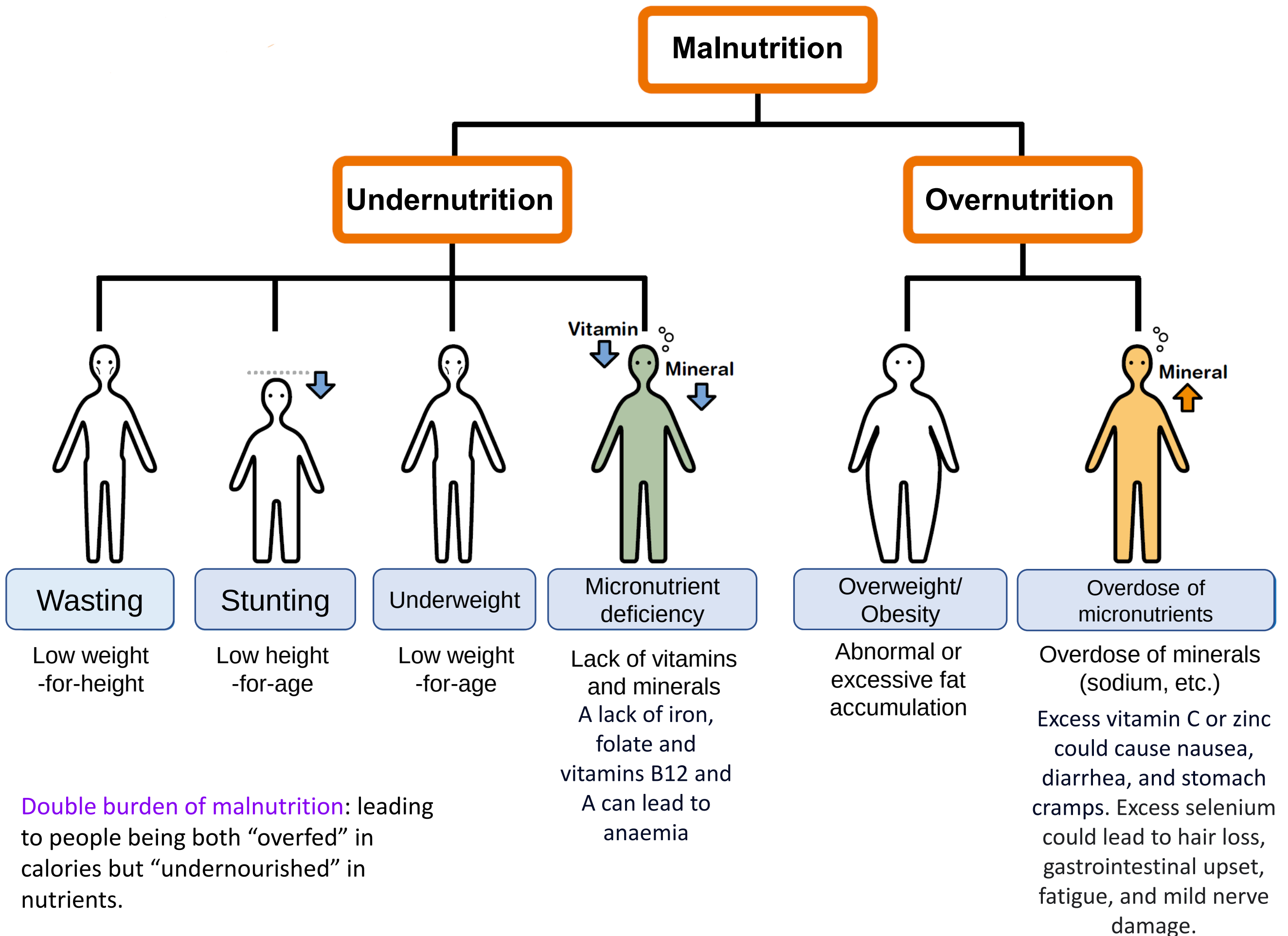
For example, if you are 25 years old, are highly active, and weigh 120 pounds, you will require $120 \times 20 = 2,400$ kilocalories daily to maintain weight. If you want to gain weight you will require more; to lose, you will require less. The amount is only a rough estimate. Other factors, such as height, must be considered. A person 5 feet, 2 inches tall and active does not require as much energy as an active 6-footer whose body weight is the same.

Malnutrition

- Malnutrition refers to **insufficient, excessive, or imbalanced** consumption of essential nutrients or impaired nutrient utilization by an organism/individual.
- In **developed countries**, the diseases of malnutrition are most often associated with **nutritional imbalances or excessive consumption**.
- The **double burden of malnutrition** consists of both undernutrition and overweight and obesity, as well as diet-related noncommunicable diseases.

❑ Malnutrition can lead to:

1. short- and long-term health problems
2. slow recovery from wounds and illnesses
3. a higher risk of infection



Some signs and **symptoms of malnutrition** include:

- weight loss
- a lack of appetite or interest in food or drink
- tiredness and irritability
- an inability to concentrate
- always feeling cold
- depression
- loss of fat, muscle mass, and body tissue
- a higher risk of getting sick and taking longer to heal
- longer healing time for wounds

Eventually, a person may also experience heart failure.

Mental health conditions: Undernutrition or malnutrition can affect people with -

- depression
- dementia
- schizophrenia
- anorexia nervosa

